

CS202: Software Tools and Techniques for CSE

Lecture 12

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January 2025

Let us Develop a Windows Forms application that accepts employee details including **Emp_Id**, **Emp_Salary**, and **Emp_Name**. Incorporate a **MenuStrip** into the form, offering four functionalities: **Add**, **Search**, **Delete**, and **Exit**.

Ensure that ***no business logic is written directly in the form***. Instead, create two separate classes:

- **Employee** – This class should include three **private fields**: **_Id**, **_Salary**, and **_Name**, along with corresponding **public properties** to access and modify these fields.
- **EmployeeManager** – This class will manage a **collection** of employees using an **ArrayList**. It should provide four public methods: **AddEmployee**, **SearchEmployee**, **DeleteEmployee**, and **GetAllEmployees** (if needed).

When a user selects a menu item, the corresponding method in the EmployeeManager class should be invoked. Selecting the **Exit** option should close the application.

Note: Duplicate Employee IDs are not permitted.

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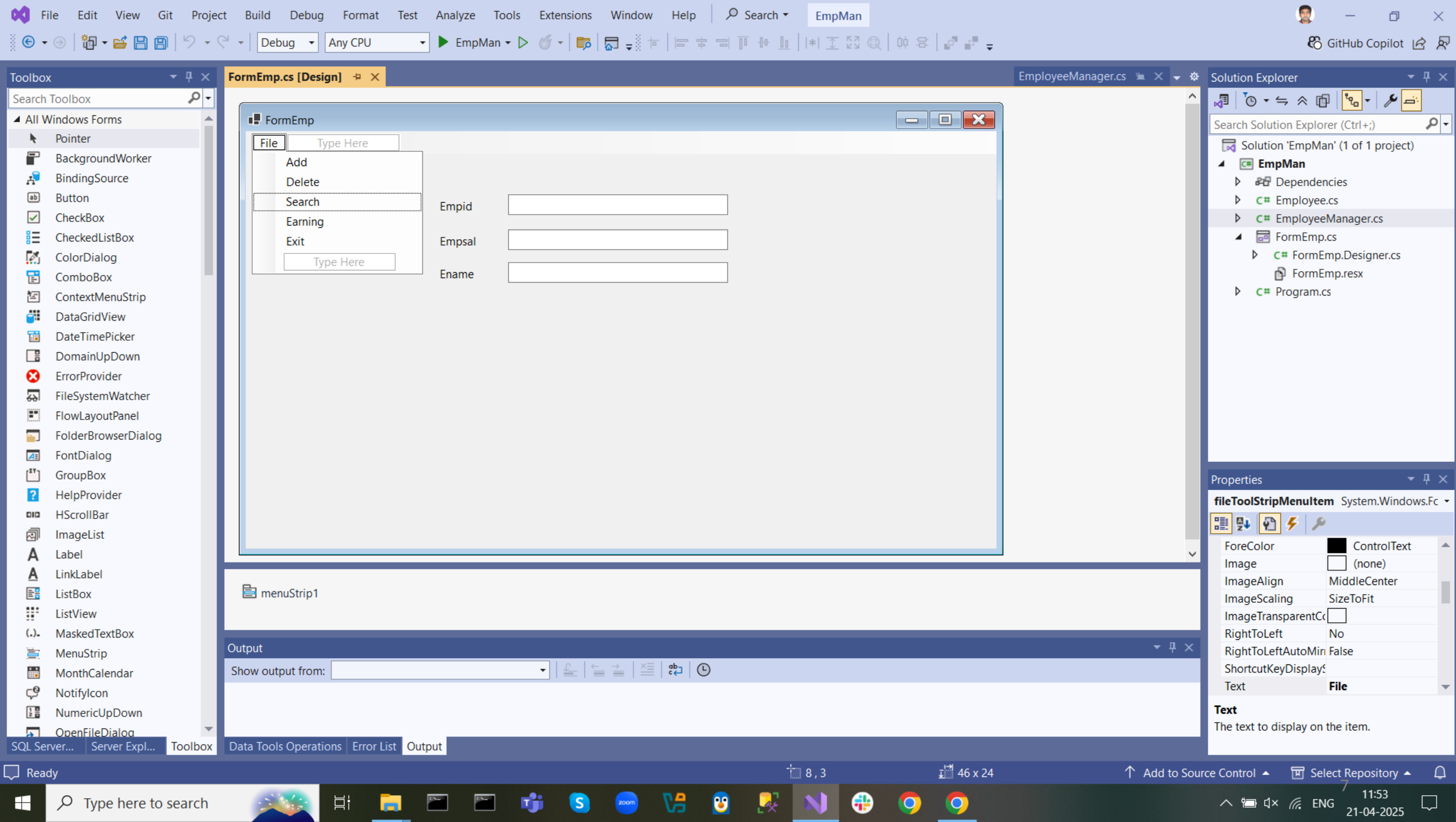
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FileEditViewGitProjectBuildDebugTestAnalyzeToolsExtensionsWindowHelp

SearchEmpMan

GitHub Copilot

DebugAny CPUEmpMan

Toolbox

Search Toolbox

All Windows Forms

Pointer

BackgroundWorker

BindingSource

Button

CheckBox

CheckedListBox

ColorDialog

ComboBox

ContextMenuStrip

DataGridView

DateTimePicker

DomainUpDown

ErrorProvider

FileSystemWatcher

FlowLayoutPanel

FolderBrowserDialog

FontDialog

GroupBox

HelpProvider

HScrollBar

ImageList

Label

LinkLabel

ListBox

ListView

MaskedTextBox

MenuStrip

MonthCalendar

NotifyIcon

NumericUpDown

OpenFileDialog

EmployeeManager.csEmployee.csFormEmp.csFormEmp.cs [Design]

FormEmp

FileType Here

Items Collection Editor

Select item and add to list below:

MenuItemAdd

Members:

menuStrip1fileToolStripMenuItem

MenuStripmenuStrip1

Behavior

AllowDropFalse

AllowItemReorderFalse

AllowMergeTrue

ContextMenuStrip(none)

EnabledTrue

ImeModeNoControl

MdiWindowListItem(none)

ShowItemToolTipsFalse

TabIndex0

TabStopFalse

VisibleTrue

Data

(DataBindings)(ControlBindings)

Tag

Design

(Name)menuStrip1

GenerateMemberTrue

OKCancel

Output

Show output from: Windows Forms

[12:18:58.797390] warn: Could not find field, 'fileToolStripMenuItem'.

[12:19:26.745897] warn: Could not find field, 'toolStripContainer1'.

Solution Explorer

Search Solution Explorer (Ctrl+;)

Solution 'EmpMan' (1 of 1 project)

EmpMan

Dependencies

Employee.cs

EmployeeManager.cs

FormEmp.cs

FormEmp.Designer.cs

FormEmp.resx

Program.cs

Program

Properties

menuStrip1System.Windows.Forms.MenuS

VisibleTrue

Data

(DataBindings)(ControlBindings)

Items(Collection)

Tag

Embed in ToolStripContainer, Insert Standard Items, Edit Items...

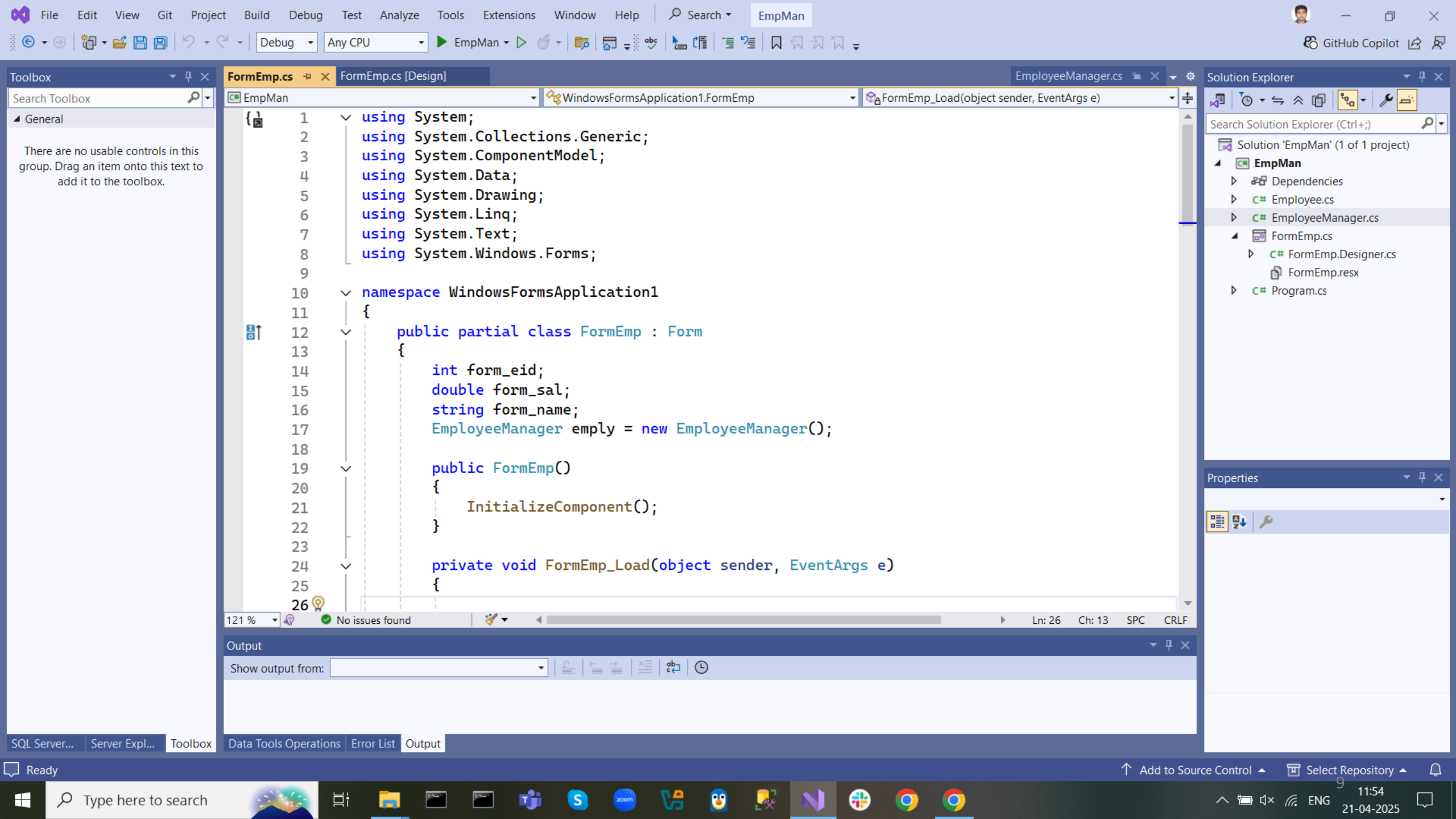
Items

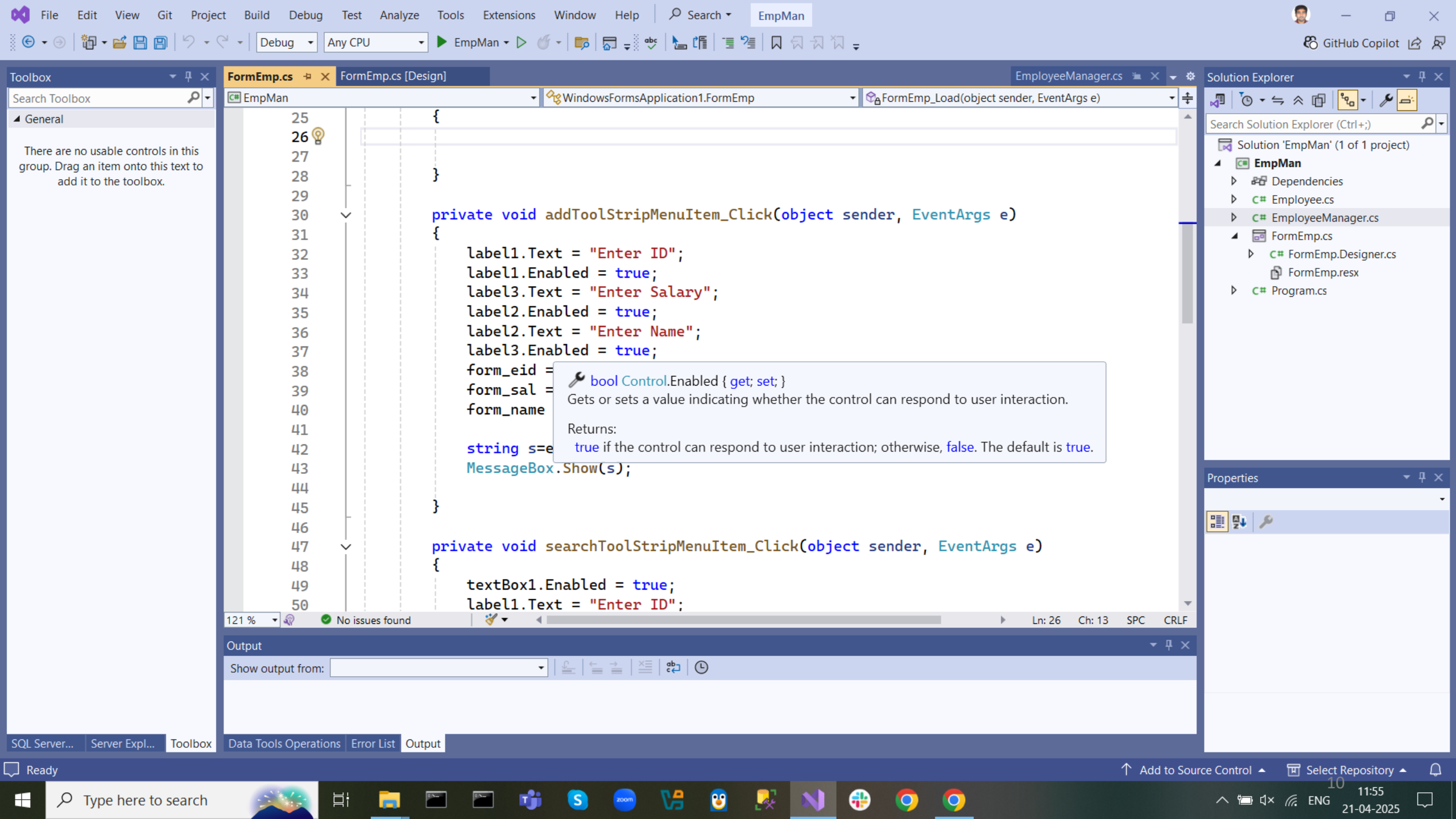
Collection of items to display on the ToolStrip.

Ready

Type here to search

12:2021-04-2025





Solution Explorer

Search Solution Explorer (Ctrl+J)

Search Solution Explorer (Ctrl+,)

Solution 'EmpMan' (1 of 1 project)

EmpMan

► Dependencies

► C# Employee.cs

EmployeeManager.cs

FormEmp.cs

- FormEmps
- C# FormEmp Designer.cs

- FormEmp.Designer.cs
- FormEmp.resx

FormEmp.resx

Program.cs

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Properties



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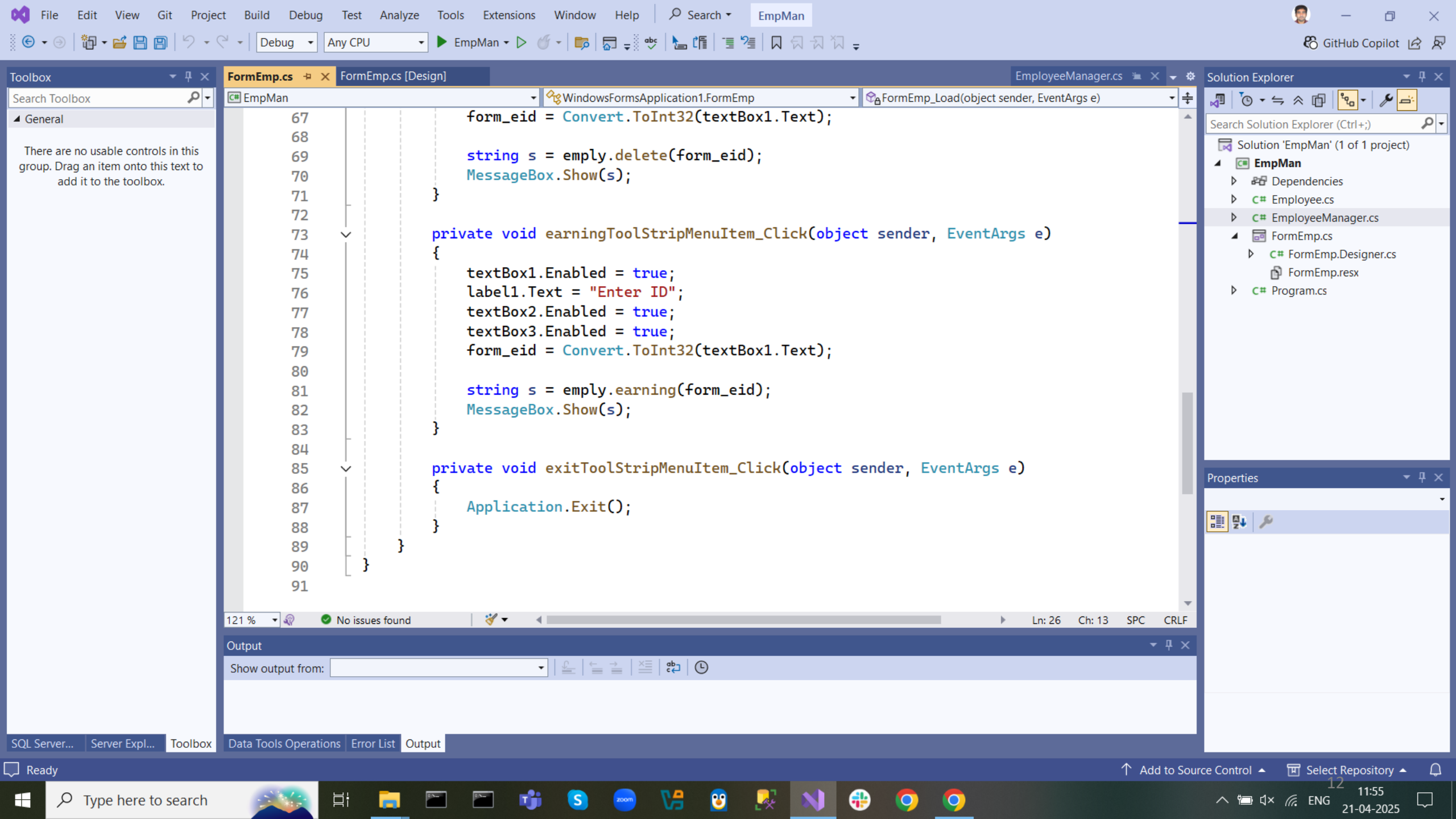
100% of the respondents in the study were female, and 90% were aged 18 years or older. The majority of the respondents were from the United States (60%), followed by Canada (20%), and the United Kingdom (10%). The remaining 10% of respondents were from various other countries, including Australia, India, and the Netherlands.

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Source Control ▲  Select Repository ▲ 

11:55

21-04-2025



The screenshot shows the Visual Studio IDE with the following details:

- Top Bar:** Displays the current file as `FormEmp.cs [Design]`. Other open files include `Program.cs`, `FormEmp.cs`, and `EmployeeManager.cs`.
- Explorer:** Shows the project structure with `EmpMan` as the selected item.
- Code Editor:** Contains the following C# code:

```
1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Windows.Forms;
5
6 namespace WindowsFormsApplication1
7 {
8     static class Program
9     {
10         /// <summary>
11         /// The main entry point for the application.
12         /// </summary>
13         [STAThread]
14         static void Main()
15         {
16             Application.EnableVisualStyles();
17             Application.SetCompatibleTextRenderingDefault(false);
18             Application.Run(new FormEmp());
19         }
20     }
21 }
```
- Bottom Bar:** Shows the status bar with `121 %` zoom, `No issues found`, and line/column coordinates `Ln: 1 Ch: 1`.
- Output Window:** The `Output` tab is active, showing `Show output from:` and various icons for output management.

The image shows two overlapping windows from the Visual Studio IDE. The top window is the 'Solution Explorer', which displays a project named 'Solution 'EmpMan' (1 of 1 project)'. Under this project, there is a folder icon and the name 'EmpMan'. Below the folder, a list of files is shown: 'Dependencies' (with a folder icon), 'Employee.cs' (with a C# icon), 'EmployeeManager.cs' (with a C# icon), 'FormEmp.cs' (with a form icon), and 'Program.cs' (with a C# icon). The 'Program.cs' file is currently selected and highlighted. The bottom window is the 'Properties' window, which is currently empty and shows a toolbar with icons for Solution Explorer, Properties, and a key icon.

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Solution Explorer

Search Solution Explorer (Ctrl+)

Solution 'EmpMan' (1 of 1 project)

EmpMan

Dependencies

▷ C# Employee.cs

▶ C# EmployeeManager.cs

- FormEmp.cs
- Program.cs

Program.cs

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Properties

[illegible]

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Abstract

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 240. **Abbrev**


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11:56
ENG 21.04.2025

EmployeeManager.cs Employee.cs Program.cs FormEmp.cs FormEmp.cs [Design]

EmpMan WindowsFormsApplication1.EmployeeManager

```
using System;
```

```
using System;
```

```
3  namespace WindowsFormsApplication1
```

```
5 class EmployeeManager < Employee
```

```
7 private ArrayList an = new ArrayList();
```

```
public AmendData() {
```

[illegible][illegible]

```
14      public string earning(int form_eid)
```

```
16         int flag = 0;
```

18

[illegible]

```
22         if (a eid == form eid)
```

```
flag = 1;
```

25. Di car,

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Show output from:      

Data Tools Operations Error List Output

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Solution 'EmpMan' (1 of 1 project)

Dependencies

- Employee.cs
- EmployeeManager.cs

- FormEmp.cs
- Program.cs

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Documentation

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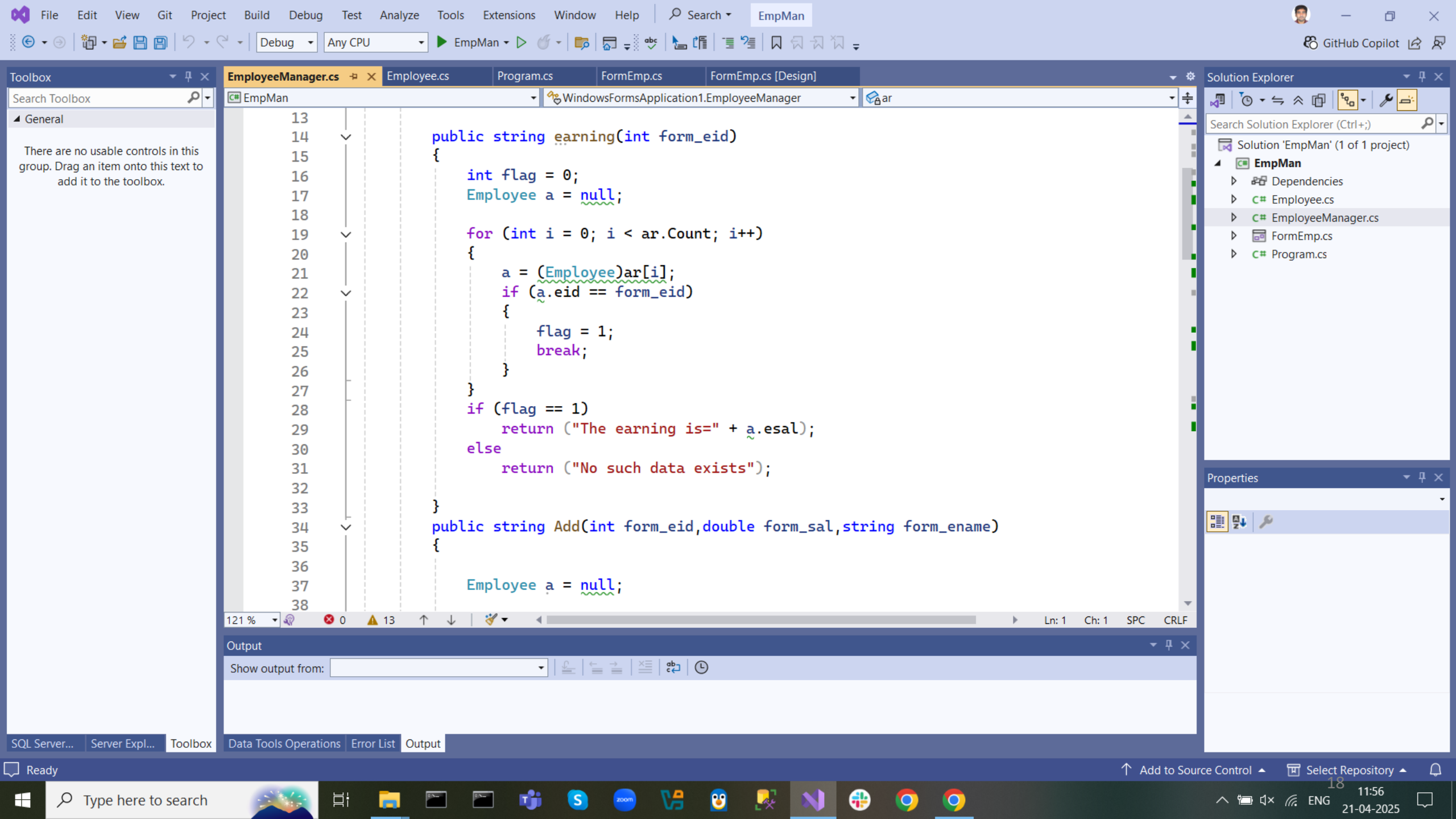
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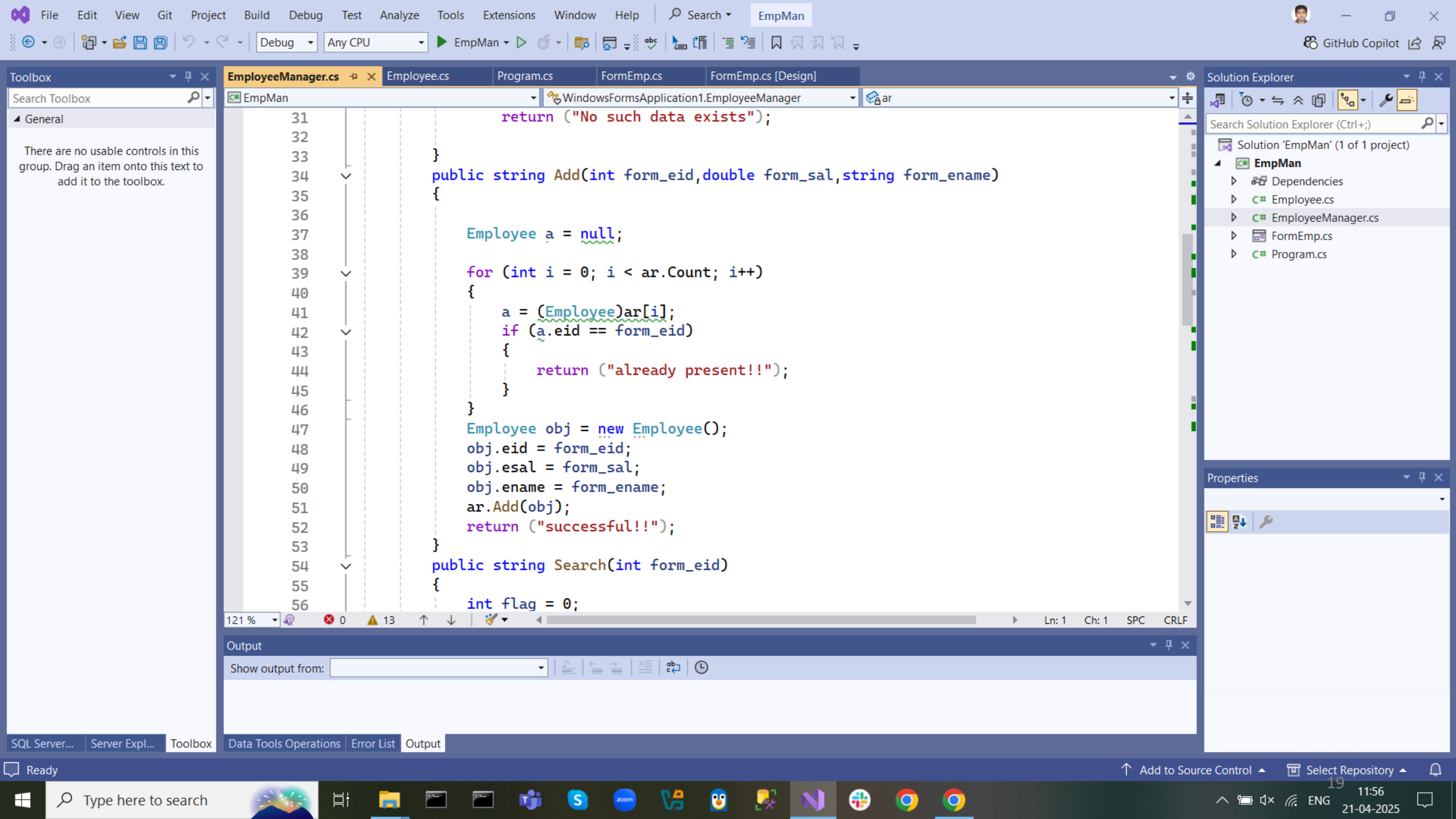
1. *Journal of the American Medical Association*, 2000; 283: 2689-2696.

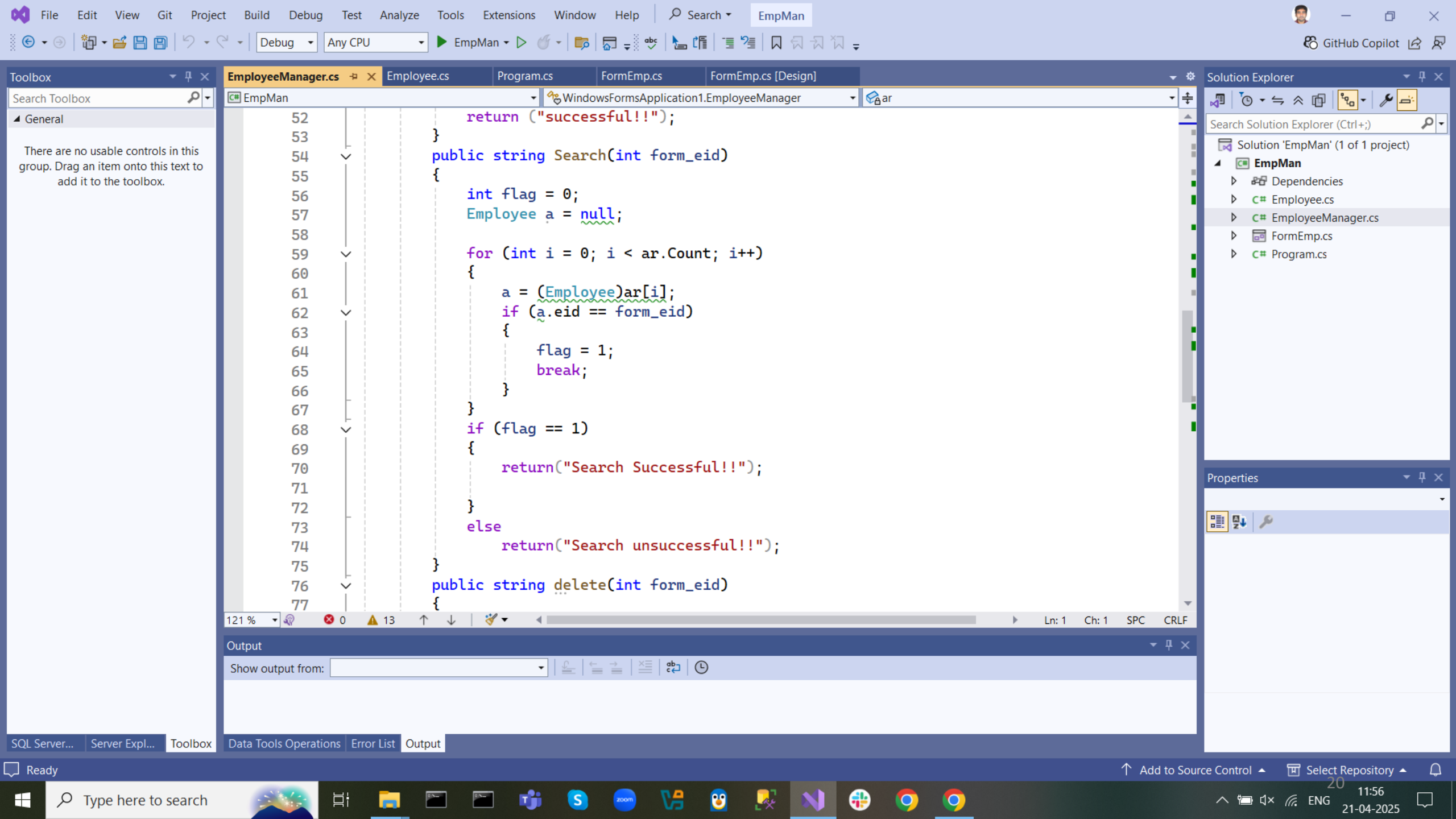
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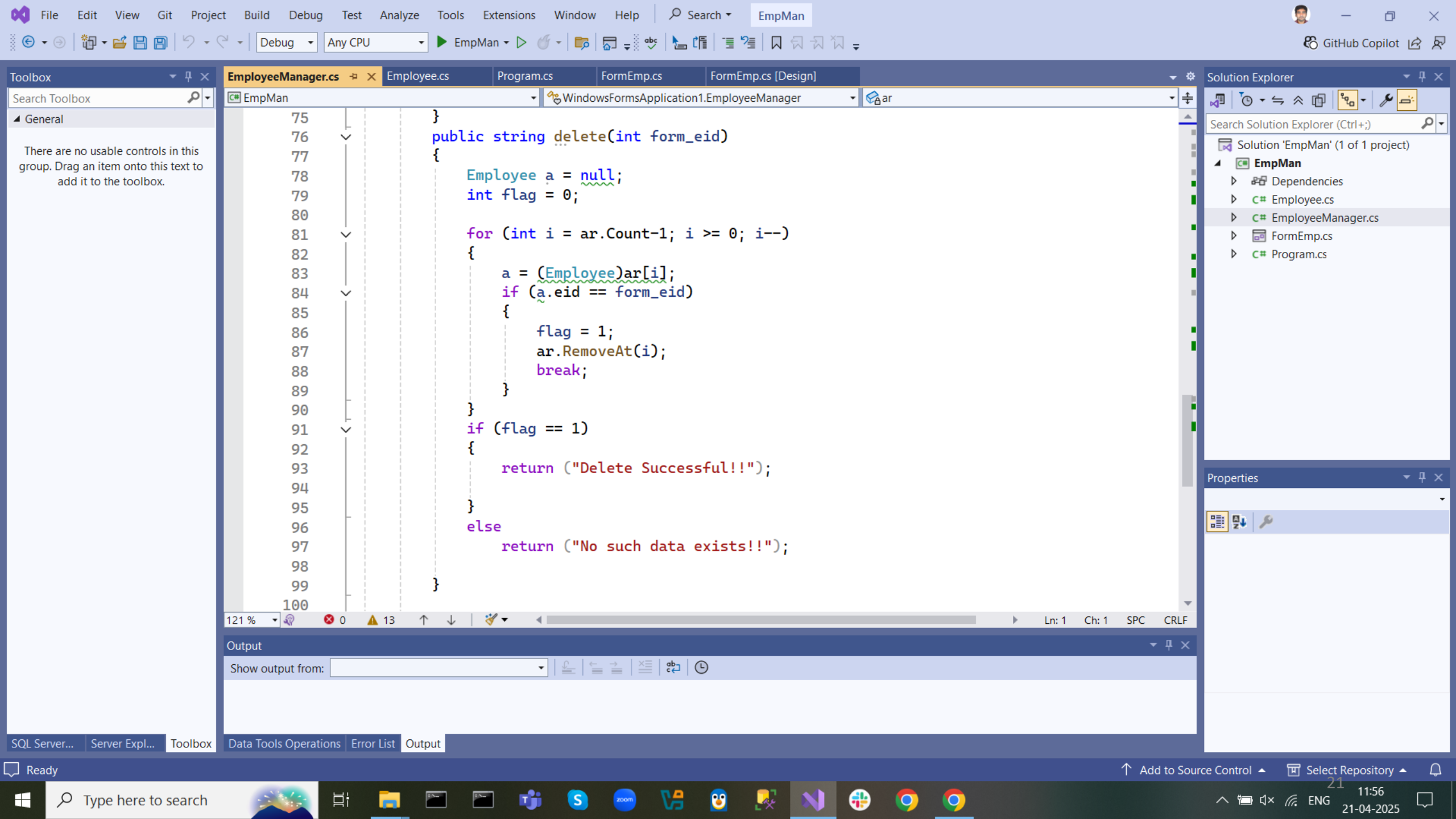
11:56
ENG 21.04.2025

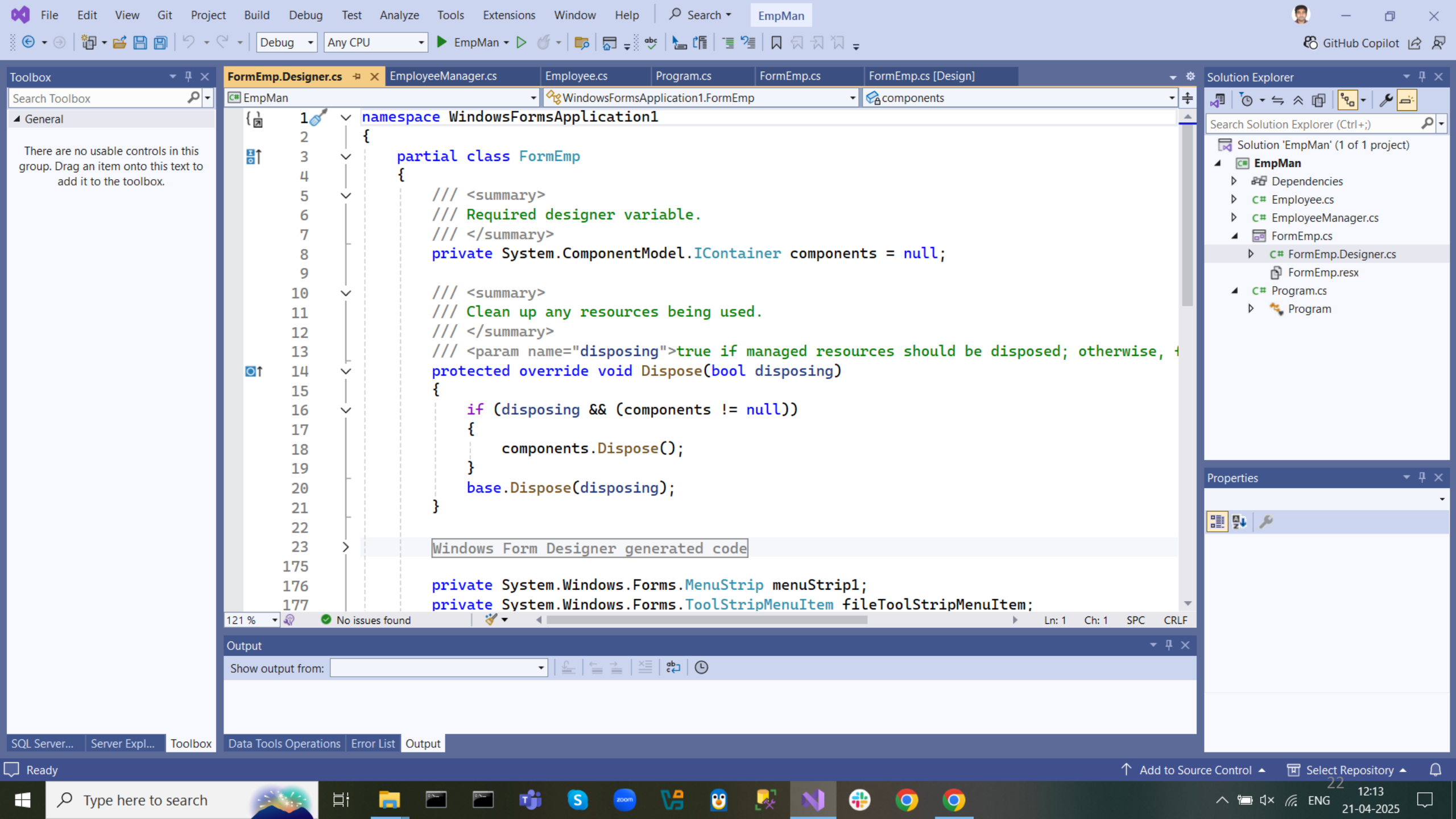
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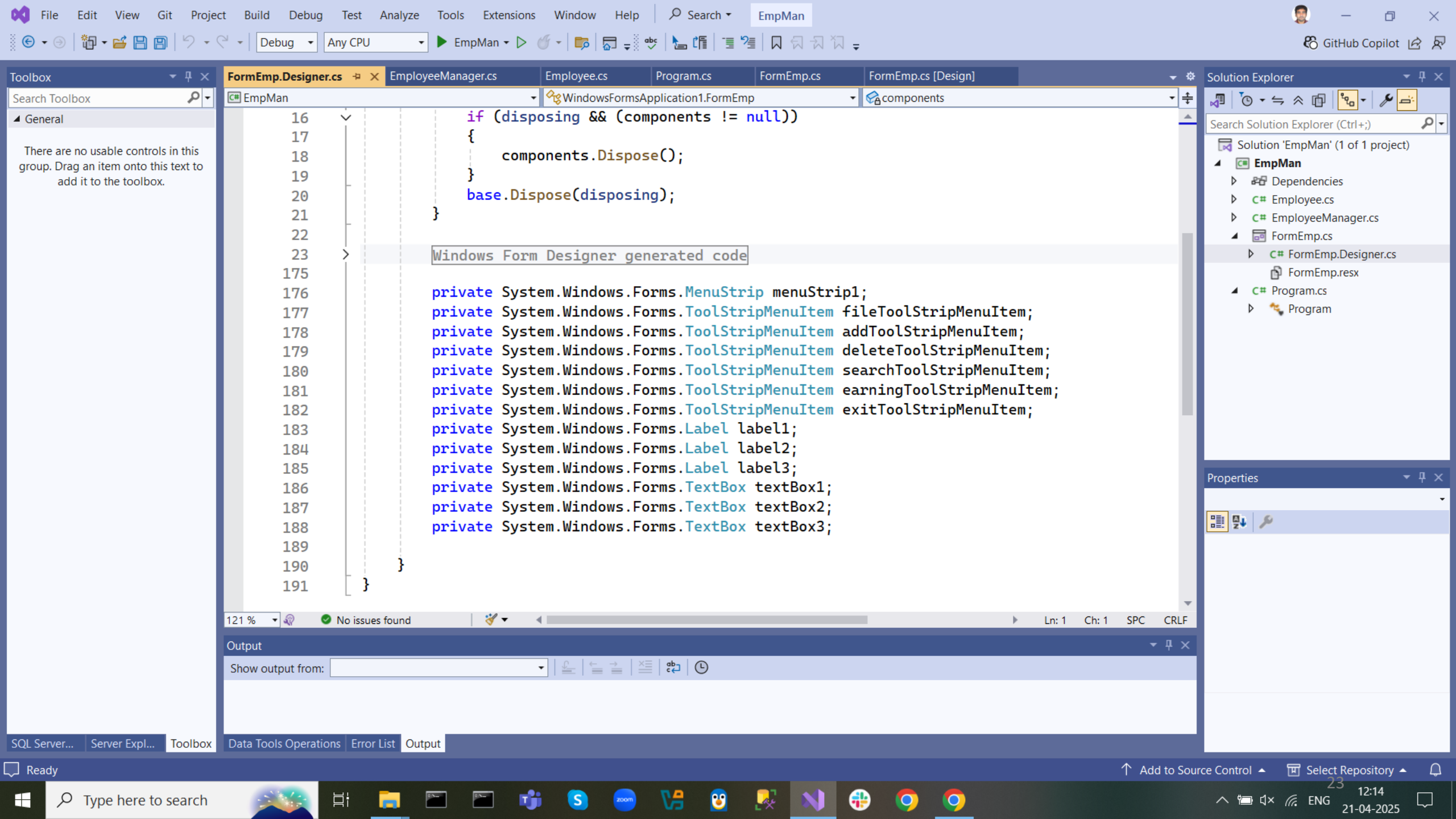












Texts, References, and Acknowledgements

Textbook:

- Sharp, J. (2022). *Microsoft Visual C# Step by Step*, 10th edition, Microsoft Press.
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